HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented Python-based **AI solutions** and automation scripts to streamline **business processes**, directly addressing repetitive tasks for a specific department.
- Developed and deployed **large language model** applications, integrating them with **business data** and APIs to solve identified operational challenges.
- Collaborated with department managers to understand their **workflows** and day-to-day challenges, translating these into clear **technical requirements** and realistic delivery plans.
- Built and evaluated **AI applications**, performing rigorous testing and gathering user feedback to assess performance, reliability, and overall usefulness.
- Engineered Python data pipelines to ingest and process **large datasets** for model training and analysis, supporting the development of **predictive models**.
- Containerized AI applications using **Docker** and deployed them to **AWS** infrastructure, ensuring secure and maintainable integration with existing systems.
- Provided practical **training and documentation** to colleagues from non-technical backgrounds, facilitating confident adoption and effective use of new **AI tools**.

SKILLS

AI/ML – Large Language Models, Prompt Design, Model Evaluation, Retrieval Augmented Generation, Embeddings, Vector Databases, AI Engineering

Programming – Python, JavaScript, TypeScript

Cloud Platforms – AWS, Azure, Google Cloud

Databases – SQL, PostgreSQL, MySQL, Oracle

Development Tools – Git, Docker, Jenkins, APIs, Testing, Debugging, Deployment

Business Systems – Workflow Automation, Business Systems Integration, Microsoft 365, SharePoint, Power Automate

PROJECTS

AI-Powered Workflow Automation for Operations

- Developed a **Python** application using **LangChain** to automate a key operational reporting workflow, reducing manual data compilation time by approximately **30%**.
- Integrated the solution with **PostgreSQL** databases and external APIs to retrieve and process relevant business data for the AI models.
- Designed and implemented prompt templates for a **large language model** to generate structured summaries of operational incidents, improving report clarity for non-technical stakeholders.

Intelligent Data Extraction for Service Improvement

- Built a **Python** script to extract and structure information from unstructured service logs using NLP techniques, feeding into a **vector database** for enhanced search capabilities.
- Developed a REST API using **FastAPI** to serve model predictions and processed data, enabling integration with existing business systems.
- Containerized the service with **Docker** and deployed it on **AWS Lambda**, ensuring scalable and efficient processing of ~500 requests per hour.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)